Evanns Morales-Cuadrado

Robotics PhD Candidate | U.S. Citizen | Fulbright Specialist
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SUMMARY

Robotics PhD candidate in Dr. Samuel Coogan's FACTS Lab at Georgia Tech, Fulbright Specialist, and year-round AI for Autonomy intern at Sandia National Laboratories. Specializes in safe autonomy, reinforcement learning, and real-time control. Proven track record of developing and deploying safe, computationally efficient planning and control methods for autonomous systems, with work published in leading peer-reviewed robotics and controls venues. Selected to the U.S. Department of State's Fulbright Specialist Roster after spearheading an international research collaboration with a Brazilian partner institution and securing research funding for the work.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

EDUCATION

Georgia Institute of Technology 2022-Present
Ph.D. in Robotics; Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington 2018-2022
Honors Bachelor of Science in Electrical Engineering; Minor in Mathematics; Minor in Physics; Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Sandia National Laboratories, Albuquerque, NM May 2026 - Present
Year-Round Graduate Research Intern, AI for Autonomy

- Developed novel reinforcement learning-based quadrotor-vs-quadrotor interception strategies, reaching targets 2.4x sooner with 14.7x more impact energy than baselines
- Created a novel Newton-Raphson (NR)-based interception law, raising interception success from 10% to 70% over standard NR
- Introduced novel benchmarking for the terminal characteristics of quadrotor interceptions
- Built C#/.NET networking frameworks and execution scripts that coordinate multi-robot communication and task execution

Georgia Institute of Technology, Atlanta, GA August 2022 - Present
Graduate Research Assistant, FACTS Lab (Advisor: Dr. Samuel Coogan)

- Deployed a novel Newton-Raphson-based tracking control law on a quadrotor, cutting error 59% on average versus PX4's native tracker
- Outperformed NMPC on a quadrotor (5x faster computation, 3.6x less CPU energy) and a blimp (65x faster computation, 16x less CPU energy), with competitive quadrotor accuracy and 2.2x lower blimp error
- Created RTD-RAX, a receding-horizon safe trajectory planner that plans less conservatively than baselines and stays safe in high-disturbance scenarios where baselines crash
- Optimized and deployed Gaussian-process runtime assurance on a quadrotor for a 100 Hz control loop, landing safely in unknown wind from an industrial fan
- Built the lab's real-time flight-test stack, enabling repeatable hardware validation

Georgia Institute of Technology, Atlanta, GA January 2024 - May 2026
Teaching Assistant, Vertically Integrated Project in Robotics

- Mentored 20 undergraduates per semester in hands-on robotics research, several of whom went on to robotics PhD programs
- Designed and delivered undergraduate lectures on ROS, Git/GitHub, Linux, and control theory, preparing students for research on robotics and autonomous control systems

University of Texas at Arlington, Arlington, TX November 2021 - May 2022
Research Assistant, Autonomous Systems Lab (Advisor: Dr. Frank Lewis)

- Trained in one of the most-cited control research groups in electrical engineering, with Dr. Lewis' work ranked 12th in the USA and 23rd worldwide by citation impact
- Contributed to reinforcement learning-based quadrotor control research

University of Texas at Arlington, Arlington, TX January 2022 - May 2022
Teaching Assistant, Graduate-Level Intelligent Systems Course

- Selected as an undergraduate teaching assistant for a graduate-level course
- Led exam review sessions and graded homework, exams, and projects

University of Texas at Arlington, Arlington, TX October 2019 - May 2022
Research Assistant, Dynamical Networks and Control Lab (Advisor: Dr. Yan Wan)

- Completed UTA Honors College thesis on learning-based minimum-time and minimum-energy multi-vehicle path planning

RESEARCH INTERESTS

- Safe Autonomy
- Hardware Deployment
- Unmanned Ground and Aerial Vehicles
- Advanced Nonlinear Control
- Learning-Based Control
- Trajectory Generation and Planning

SELECTED PUBLICATIONS

E. Morales-Cuadrado, S. Coogan, Z. Kakish, and A. Parikh. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA). (title redacted until the double-blind review process is complete)*

L. Pinto*, **E. Morales-Cuadrado***, S. Coogan, M. A. Wehrmeister, L. Caselatto, and C. E. Magrin. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA). *Equal contribution. (title redacted until the double-blind review process is complete)*

M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, "Trajectory Tracking for Systems with Unknown Time-Varying Disturbances." *IEEE Open Journal of Control Systems (OJ-CSYS)*, 2026.

E. Morales-Cuadrado, "Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance." Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. *pdf available. slides available.*

E. Morales-Cuadrado, L. Chung, S. Kousik, and S. Coogan, "`RTD-RAX`: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances." Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). *(accepted to both) preprint available.*

E. Morales-Cuadrado, L. Baird, Y. Wardi, and S. Coogan, "Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method." *IEEE Transactions on Control Systems Technology (TCST)*, 2026. *preprint available.*

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, "Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic." Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. *(under revision) preprint available.*

E. Morales-Cuadrado, C. Llanes, Y. Wardi, and S. Coogan, "Newton-Raphson Flow for Aggressive Quadrotor Tracking Control." *2024 American Control Conference (ACC). preprint available.*

OPEN-SOURCE CONTRIBUTIONS

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech's Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

SERVICE & LEADERSHIP

Georgia Institute of Technology, Atlanta, GA August 2024 - Present
Co-Founder and President of Puerto Rican Student Association

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

University of Texas at Arlington, Arlington, TX August 2021 - May 2022
Undergraduate Research Mentor

- As the lead undergraduate student in Dr. Frank Lewis' Autonomous Systems Lab, mentored junior undergrads in controls research and guided them on a research project in ground vehicle control
- Guided students in Robot Operating System, Python, Linux, and Control Systems

University of Texas at Arlington, Arlington, TX August 2020 - May 2022
Eta Kappa Nu Public Relations Chair & Circuits I Tutor

- Executive board member of UT Arlington's IEEE-HKN chapter
- In charge of social media advertising for the honor society's events and outreach initiatives for the electrical engineering student body
- Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I

HONORS AND AWARDS

Fulbright Specialist August 2026
Peer-review-selected to the U.S. Department of State's Fulbright Specialist Roster for a three-year term

Top-3 Finalist at a Deep Learning Research Symposium November 2023
Professor-sponsored award for novel research in deep learning and robotics

Goizueta Fellowship at Georgia Tech August 2022
Highly selective fellowship for graduate students of Latin-American descent

Summa Cum Laude Honors at University of Texas at Arlington May 2022
Exceptional academic distinction in top GPA tier for graduation honors

Honors College Degree Requirements at University of Texas at Arlington December 2021
Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee

Electrical Engineering Honors Scholar at University of Texas at Arlington December 2021
Inaugural member of the electrical engineering honors cohort

IEEE-HKN International Electrical Engineering Honor Society Membership August 2021
Selective undergraduate society recognizing outstanding academic achievement and community service

Research Experience for Undergraduates Fellowship Recipient August 2020
Fellowship to sponsor promising young undergraduate researchers

Chance Vought Engineering and Science Endowment Scholarship August 2020
Highly selective award given annually to three recipients across the entire College of Engineering

Toray Composite Materials America Inc. Sons and Daughters Scholarship May 2019
Academic scholarship awarded to children of employees of Toray America

James E. Casey Scholarship Recipient January 2019
Academic scholarship awarded to children of employees of UPS

James R. Hoffa Memorial Scholarship Recipient May 2018
Academic scholarship awarded to children of Teamsters Union Members

Grapevine, TX Rotary Club Scholarship Recipient May 2018
Local rotary club scholarship for academic and civically-minded scholars

National Hispanic Scholar March 2018
Recognition of top national performers of Latin descent on the SAT

National Merit Scholar March 2018
Recognition of top national performers on the SAT